

# Edward Turk

Environmental Scientist | GIS & Data Analysis

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## Summary

Environmental scientist completing an M.S. in Environmental Management at the University of San Francisco (expected August 2026), with an Advanced GIS Certificate. Applies ArcGIS Pro, Google Earth Engine, R, and Python to floodplain hydrology and vegetation analysis. Brings five years of applied scientific work across academic, ag-tech, and biotech settings, including data management, laboratory quality control, and technical documentation, with first-author research presented at AGU 2025.

## Core Skills

**GIS & Remote Sensing:** ArcGIS Pro, Google Earth Engine, QGIS, Sentinel-2 imagery, LiDAR, land surface phenology, spectral indices, DEM/terrain analysis (HAND, flow accumulation, watershed delineation), kriging and spatial interpolation

**Programming & Data:** R (mixed-effects models, geostatistics, visualization), Python (ArcPy, numpy, rasterio), JavaScript (Earth Engine), Jupyter, basic SQL, geodatabases, spatial data QA/QC and metadata standards

**Field & Sampling:** Water, soil, and porewater sampling, groundwater monitoring, gas flux measurement, handheld GPS coordinate collection, lab coordination

**Quality Control & Communication:** ISO-aligned QA/QC, method validation, SOP development, technical writing, data visualization, presentations to stakeholders and scientific advisors

## Education

**M.S. Environmental Management**, *University of San Francisco* **2024 - 2026**

- Recipient of the Joseph Petulla Award for the Outstanding Environmental Management Graduate Student, MSEM Class of 2026.
- Advanced GIS Certificate, earned through the MSEM program.
- Relevant coursework: Remote Sensing; LiDAR; Google Earth Engine; Python for GIS; Advanced Data Analysis in R; Applied GIS.

**B.S. Ecology and Evolution**, *University of California, Santa Cruz* **2014 - 2018**

## Selected Projects

**Master's Thesis: Complementing Topographic Wetness with Sentinel-2 Phenological Indices to Map Soil Moisture Gradients in a Snowmelt-Driven Mountain Floodplain**, *University of San Francisco*

- Modeled soil water content from Sentinel-2 phenology and a LiDAR-derived topographic wetness index using R mixed-effects models, validated by leave-one-site-out cross-validation and a held-out-year transfer test; built the Python/Google Earth Engine pipeline behind it, fitting per-pixel Fourier harmonic regressions to 419 cloud-filtered scene-dates across a ~6,000 km<sup>2</sup> watershed.
- Delivered 10 m annual water-content rasters (2018 - 2025) and wetness-class maps; basis of a first-author manuscript in preparation for the *Journal of Remote Sensing*.

**Central Valley Winter Cover-Crop Map**, *Google Earth Engine | Sentinel-2 | Python*

- Mapped winter cover-crop adoption and density in California almond orchards across six winters using SAVI and BSI indices over eight fixed calendar periods per winter, on orchard boundaries from USDA Crop Sequence Boundaries filtered by Cropland Data Layer almond classification.
- Delivered 16,365 classified orchards across 9 counties, scored for intensity, fill, and uniformity with k-means season-behavior clustering, published in a self-built web map.

## Professional Experience

**Graduate Research Assistant**, *University of San Francisco* **Feb 2025 - Present**

- Conducted floodplain field campaigns for a Department of Energy-funded methane study, including porewater chemistry sampling, groundwater depth monitoring, and methane flux measurement.
- Planned sampling logistics and coordinated data exchange across USF, Stanford SLAC, and Cal State East Bay researchers.
- Analyzed flux, porewater, and terrain data in R and Python; presented results as a first-author poster at AGU 2025, with a first-author methods manuscript in preparation.

**Quality Associate II**, *ZymoChem* **Jan 2024 - Nov 2024**

- Led quality-control testing for the R&D of a biodegradable super absorbent polymer (SAP), validating assays against ISO 17190 and sourcing national reference materials.

**Research Associate II**, *Pivot Bio* **Aug 2020 - Oct 2023**

- Technical lead for biological seed-treatment formulation research and development.
- Designed experiments, prepared invention disclosures with patent counsel, and presented results to interdisciplinary teams and to a grower audience at an agricultural conference.

**Junior Specialist**, *University of California, Davis* **Jan 2019 - Aug 2020**

- Assisted in the production of the CDFA-FREP nitrogen-removal values that feed the Central Valley Irrigated Lands Regulatory Program, reported to the Central Valley Regional Water Quality Control Board.